Chapter 6. Metrization Theorems and Paracompactness

Local finiteness, Nagata–Smirnov, paracompactness, Smirnov metrization

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6.1 Local finiteness

Motivation. Urysohn metrization needs a countable basis; to go further we need a substitute. A locally finite family of sets, one where each point meets only finitely many of them, is the right weakening, since unions and continuous functions defined piece by piece behave well for such families.

A collection $\mathcal{A}$ of subsets of X is **locally finite** if every point has a neighbourhood meeting only finitely many elements of $\mathcal{A}$; it is **countably locally finite** (σ-locally finite) if it is a countable union of locally finite collections. A collection $\mathcal{B}$ **refines** $\mathcal{A}$ (is a **refinement** of $\mathcal{A}$) if each $B \in \mathcal{B}$ lies in some $A \in \mathcal{A}$; it is an open (closed) refinement if its elements are open (closed).

Lemma 6.1 (39.1)

Let $\mathcal{A}$ be locally finite.
(a) Every subcollection of $\mathcal{A}$ is locally finite.
(b) $\{\overline{A} : A \in \mathcal{A}\}$ is locally finite.
(c) $\overline{\bigcup_{A\in\mathcal{A}} A} = \bigcup_{A\in\mathcal{A}} \overline{A}$.

Proof. (a) is clear.

(b) An open set U meeting $\overline{A}$ meets A; so a neighbourhood meeting finitely many A meets only the corresponding $\overline{A}$.

(c) The inclusion $\supset$ always holds. Let $x \in \overline{\bigcup A}$ and let U be a neighbourhood of x meeting only $A_1, \dots, A_k$. If $x \notin \overline{A_1} \cup \cdots \cup \overline{A_k}$, then $U - (\overline{A_1} \cup \cdots \cup \overline{A_k})$ is a neighbourhood of x missing every element of $\mathcal{A}$, a contradiction. ■

Without local finiteness (c) fails: the one-point sets $\{1/n\}$ have union with closure containing 0.

Lemma 6.2 (39.2)

If X is metrizable, every open covering $\mathcal{A}$ of X has a countably locally finite open refinement $\mathcal{E}$ that covers X.

Proof. Fix a metric d and a well-ordering $<$ of $\mathcal{A}$ (Theorem 1.25). For $n \in \mathbb{Z}_+$ and $U \in \mathcal{A}$ let

$$S_n(U) = \{x : B(x, 1/n) \subset U\}, \qquad T_n(U) = S_n(U) - \bigcup_{V<U} V, \qquad E_n(U) = \bigcup_{x\in T_n(U)} B(x, 1/3n).$$

If $V < U$, $x \in T_n(V)$, $y \in T_n(U)$, then $B(x, 1/n) \subset V$ and $y \notin V$, so $d(x, y) \geq 1/n$. Hence distinct sets $E_n(U), E_n(V)$ are at distance at least $1/3n$, a ball of radius $1/6n$ meets at most one of them, and $\mathcal{E}_n = \{E_n(U)\}_U$ is locally finite. Also $E_n(U) \subset U$, since $B(x, 1/3n) \subset B(x, 1/n) \subset U$ for $x \in T_n(U)$. Finally, given x, let U be the first element of $\mathcal{A}$ containing x and choose n with $B(x, 1/n) \subset U$: then $x \in S_n(U)$ and x lies in no $V < U$, so $x \in T_n(U) \subset E_n(U)$. Take $\mathcal{E} = \bigcup_n \mathcal{E}_n$. ■

6.2 The Nagata–Smirnov metrization theorem

Motivation. This gives the complete answer to the metrization problem: a space is metrizable if and only if it is regular with a countably locally finite basis. The proof follows the Urysohn embedding, with a countable basis replaced by a countably locally finite one.

A set is a G_δ **set** if it is a countable intersection of open sets. In a metric space every closed set A is G_δ: $A = \bigcap_n \{x : d(x, A) < 1/n\}$.

Lemma 6.3 (40.1)

Let X be regular with a countably locally finite basis $\mathcal{B}$. Then X is normal and every closed set is a G_δ set.

Proof.

Step 1. for each open W there are open $U_1, U_2, \dots$ with $W = \bigcup U_n = \bigcup \overline{U_n}$. Write $\mathcal{B} = \bigcup \mathcal{B}_n$ with $\mathcal{B}_n$ locally finite, let $\mathcal{C}_n = \{B \in \mathcal{B}_n : \overline{B} \subset W\}$ (locally finite) and $U_n = \bigcup_{B \in \mathcal{C}_n} B$. By Lemma 6.1(c), $\overline{U_n} = \bigcup_{\mathcal{C}_n} \overline{B} \subset W$. If $x \in W$, regularity gives a basis element $B \ni x$ with $\overline{B} \subset W$; $B \in \mathcal{C}_n$ for some n, so $x \in U_n$.

Step 2. if C is closed, apply Step 1 to $W = X - C$: $C = \bigcap_n (X - \overline{U_n})$, a G_δ.

Step 3. for disjoint closed C, D, Step 1 with $W = X - D$ gives open U_n covering C with closures missing D, and with $W = X - C$ open V_n covering D with closures missing C. Now finish exactly as in the proof of Theorem 4.6. ■

Lemma 6.4 (40.2)

If X is normal and A is a closed G_δ set, there is a continuous $f : X \to [0, 1]$ with $f = 0$ on A and $f > 0$ off A.

Proof. Write $A = \bigcap U_n$ with U_n open. Urysohn gives $f_n : X \to [0, 1]$ equal to 0 on A and 1 on $X - U_n$. Let $f = \sum_n f_n / 2^n$, continuous by uniform convergence (Theorem 2.42). It vanishes on A; if $x \notin A$, then $x \notin U_n$ for some n, so $f(x) \geq 2^{-n} > 0$. ■

Theorem 6.5 (40.3, Nagata–Smirnov metrization theorem)

X is metrizable iff X is regular and has a countably locally finite basis.

Proof. ($\Leftarrow$)

Step 1. let $\mathcal{B} = \bigcup \mathcal{B}_n$ with $\mathcal{B}_n$ locally finite. X is normal and $X - B$ is a closed G_δ for each $B \in \mathcal{B}$ (Lemma 6.3), so Lemma 6.4 gives $f_{n,B} : X \to [0, 1/n]$ positive on B and zero off B. For $x_0 \in U$ choose $B \in \mathcal{B}_n$ with $x_0 \in B \subset U$; then $f_{n,B}(x_0) > 0$ and $f_{n,B} = 0$ off U.

Step 2. let $J = \{(n, B) : B \in \mathcal{B}_n\}$ and $F(x) = (f_{n,B}(x)) \in [0, 1]^J$; give $[0, 1]^J$ the uniform metric $\bar{\rho}$. By the proof of Theorem 4.13, F is injective and maps open sets of X to sets open in $F(X)$ for the product topology, hence also for the finer uniform topology (Theorem 2.35).

Step 3 (F is continuous into the uniform metric). let $x_0 \in X$ and $\epsilon > 0$. For each n choose a neighbourhood U_n of x_0 meeting only finitely many elements of $\mathcal{B}_n$; all but finitely many $f_{n,B}$ vanish on U_n, and for the remaining ones continuity gives a smaller neighbourhood W_n on which $|f_{n,B}(x) - f_{n,B}(x_0)| \leq \epsilon/2$ for every $B \in \mathcal{B}_n$. Choose N with $1/N \leq \epsilon/2$ and let $W = W_1 \cap \cdots \cap W_N$. For $x \in W$ the coordinates with $n \leq N$ differ by at most $\epsilon/2$, and those with $n > N$ lie in $[0, 1/n]$, so differ by at most $1/n < \epsilon/2$. Thus $\bar{\rho}(F(x), F(x_0)) \leq \epsilon/2 < \epsilon$, and F imbeds X in a metric space.

($\Rightarrow$) A metrizable space is regular (Theorem 4.7). For each m let $\mathcal{A}_m$ be the covering by balls of radius $1/m$ and $\mathcal{D}_m$ a countably locally finite open refinement covering X (Lemma 6.2); its elements have diameter at most $2/m$. $\mathcal{D} = \bigcup_m \mathcal{D}_m$ is countably locally finite. It is a basis: if $x \in U$ open, choose m with $B(x, 1/m) \subset U$ and $D \in \mathcal{D}_{3m}$ containing x; D has diameter at most $2/3m < 1/m$, so $D \subset B(x, 1/m) \subset U$. ■

6.3 Paracompactness

Motivation. Many constructions in geometry need partitions of unity subordinate to arbitrary open covers. Paracompactness, every open cover has a locally finite refinement, is precisely what makes these possible, and every metric space has it.

X is **paracompact** if every open covering has a locally finite open refinement that covers X. Compact spaces are paracompact, and so is $\mathbb{R}^n$ (Theorem 6.9 below).

Theorem 6.6 (41.1)

Every paracompact Hausdorff space is normal.

Proof. Regularity: let $a \notin B$, B closed. For each $b \in B$ choose open $U_b \ni b$ with $a \notin \overline{U_b}$ (Hausdorff). Cover X by the U_b and $X - B$, take a locally finite open refinement $\mathcal{C}$ covering X, and let $\mathcal{D}$ be the elements of $\mathcal{C}$ meeting B. Each $D \in \mathcal{D}$ is not inside $X - B$, so $D \subset U_b$ for some b and $a \notin \overline{D}$. $V = \bigcup_{\mathcal{D}} D$ is open, contains B, and $\overline{V} = \bigcup \overline{D} \not\ni a$ (Lemma 6.1). Then V and $X - \overline{V}$ separate B and a. Normality: the same argument with a closed set A in place of a, using regularity to choose U_b with $\overline{U_b} \cap A = \emptyset$. ■

Theorem 6.7 (41.2)

Every closed subspace of a paracompact space is paracompact.

Proof. Let Y be closed and $\mathcal{A}$ a covering of Y by sets open in Y; for each $A \in \mathcal{A}$ choose A' open in X with $A' \cap Y = A$. Cover X by the A' together with $X - Y$, and take a locally finite open refinement $\mathcal{B}$ covering X. Then $\{B \cap Y : B \in \mathcal{B}\}$ covers Y, is locally finite, consists of sets open in Y, and refines $\mathcal{A}$ (a B inside $X - Y$ gives the empty set). ■

A paracompact space can have non-paracompact subspaces: $\overline{S}_\Omega$ is compact, but S_Ω is not paracompact (it is not metrizable yet is locally metrizable; see Theorem 6.14). The product $\mathbb{R}^2_\ell$ of paracompact spaces is not paracompact, because it is Hausdorff but not normal.

Lemma 6.8 (41.3, Michael)

Let X be regular. The following are equivalent: every open covering of X has a refinement that is (1) an open covering of X and countably locally finite;
(2) a covering of X and locally finite;
(3) a closed covering of X and locally finite;
(4) an open covering of X and locally finite.

Proof. (4)⇒(1) is trivial.

(1)⇒(2): let $\mathcal{B} = \bigcup \mathcal{B}_n$ be an open refinement as in (1), $V_i = \bigcup_{U \in \mathcal{B}_i} U$, and for $U \in \mathcal{B}_n$ put $S_n(U) = U - \bigcup_{i<n} V_i$. The sets $S_n(U)$ refine $\mathcal{A}$ and cover X: for x, take the smallest N with x in some $U \in \mathcal{B}_N$; then $x \in S_N(U)$. Local finiteness: with $x \in U \in \mathcal{B}_N$, the open set $U \subset V_N$ meets no $S_n(V)$ with $n > N$; for each $n \le N$ choose a neighbourhood W_n meeting finitely many elements of $\mathcal{B}_n$, hence finitely many $S_n(V)$. Then $U \cap W_1 \cap \cdots \cap W_N$ meets finitely many sets $S_n(V)$.

(2)⇒(3): let $\mathcal{B}$ be the open sets U with $\overline{U}$ contained in some element of $\mathcal{A}$; it covers X by regularity. Take a locally finite refinement $\mathcal{C}$ covering X; $\{\overline{C}\}$ is locally finite (Lemma 6.1), closed, covers X and refines $\mathcal{A}$.

(3)⇒(4): let $\mathcal{B}$ be a locally finite refinement of $\mathcal{A}$ covering X (by (3)). Let $\mathcal{C}$ be the open sets meeting only finitely many elements of $\mathcal{B}$; it covers X. By (3) take a closed locally finite refinement

$\mathcal{D}$ of $\mathcal{C}$ covering X; each $D \in \mathcal{D}$ meets only finitely many elements of $\mathcal{B}$. For $B \in \mathcal{B}$ let $C(B) = X - \bigcup\{D \in \mathcal{D} : D \cap B = \emptyset\}$, open by Lemma 6.1(c) and containing B, and choose $F(B) \in \mathcal{A}$ with $B \subset F(B)$. The collection $\mathcal{E} = \{C(B) \cap F(B) : B \in \mathcal{B}\}$ is an open refinement of $\mathcal{A}$ covering X. It is locally finite: a point has a neighbourhood W meeting only $D_1, \ldots, D_k \in \mathcal{D}$, so $W \subset D_1 \cup \cdots \cup D_k$. If D meets $C(B)$, then D meets B (otherwise $D \subset X - C(B)$); hence each D_i meets only finitely many $C(B)$, and so does W. ■

Theorem 6.9 (41.4)

Every metrizable space is paracompact.

Proof. It is regular, and condition (1) of Lemma 6.8 holds by Lemma 6.2. ■

Theorem 6.10 (41.5)

Every regular Lindelöf space is paracompact.

Proof. A countable subcovering is a countable union of one-element (hence locally finite) collections, so condition (1) of Lemma 6.8 holds. ■

So $\mathbb{R}_\ell$ is paracompact, while $\mathbb{R}_\ell^2$ is not; and $\mathbb{R}^\omega$ in the product topology is paracompact (metrizable), whereas $\mathbb{R}^J$ for uncountable J is not (it is not normal).

Lemma 6.11 (41.6)

Let X be paracompact Hausdorff and $\{U_\alpha\}_{\alpha \in J}$ an indexed open covering. There is a locally finite indexed open covering $\{V_\alpha\}_{\alpha \in J}$ with $\overline{V_\alpha} \subset U_\alpha$ for each α (a **precise refinement**).

Proof. Let $\mathcal{A}$ be the open sets A with $\overline{A}$ in some U_α; it covers X by regularity (Theorem 6.6). Take a locally finite open refinement $\mathcal{B}$ covering X, and for each B choose $f(B) \in J$ with $\overline{B} \subset U_{f(B)}$. Let $V_\alpha = \bigcup\{B : f(B) = \alpha\}$. The V_α are open and cover X; a neighbourhood meeting finitely many B meets finitely many V_α; and $\overline{V_\alpha} = \bigcup_{f(B)=\alpha} \overline{B} \subset U_\alpha$ by Lemma 6.1(c). ■

Theorem 6.12 (41.7)

If X is paracompact Hausdorff and $\{U_\alpha\}_{\alpha \in J}$ is an indexed open covering, there is a partition of unity dominated by $\{U_\alpha\}$: continuous $\phi_\alpha : X \to [0,1]$ with $\operatorname{supp} \phi_\alpha \subset U_\alpha$, $\{\operatorname{supp} \phi_\alpha\}$ locally finite, and $\sum_\alpha \phi_\alpha = 1$.

Proof. By Lemma 6.11 twice, take locally finite open coverings $\{V_\alpha\}$, $\{W_\alpha\}$ with $\overline{W_\alpha} \subset V_\alpha$, $\overline{V_\alpha} \subset U_\alpha$. Urysohn (with X normal) gives ψ_α equal to 1 on $\overline{W_\alpha}$ and 0 off V_α, so $\operatorname{supp} \psi_\alpha \subset \overline{V_\alpha} \subset U_\alpha$, and these supports are locally finite (Lemma 6.1(b)). $\Psi = \sum \psi_\alpha$ is near each point a finite sum of continuous functions, hence continuous, and positive since the W_α cover. Put $\phi_\alpha = \psi_\alpha / \Psi$. ■

Theorem 6.13 (41.8)

Let X be paracompact Hausdorff, $\mathcal{C}$ a locally finite collection of subsets, and $\epsilon_C > 0$ for each $C \in \mathcal{C}$. There is a continuous $f : X \to \mathbb{R}$ with $f > 0$ everywhere and $f(x) \le \epsilon_C$ for $x \in C$.

Proof. Cover X by open sets each meeting finitely many elements of $\mathcal{C}$, and take a dominated partition of unity $\{\phi_U\}$ indexed by this covering. Let δ_U be the minimum of ϵ_C over the finitely many C meeting U ($\delta_U = 1$ if there are none), and $f = \sum_U \delta_U \phi_U$, continuous and positive. If $x \in C$ and $\phi_U(x) > 0$, then $x \in U$, so U meets C and $\delta_U \le \epsilon_C$; hence $f(x) \le \epsilon_C \sum \phi_U(x) = \epsilon_C$. ■

6.4 The Smirnov metrization theorem

Motivation. For paracompact spaces the metrization question reduces to a local one: a paracompact Hausdorff space is metrizable exactly when it is locally metrizable. This is how one shows that manifolds are metrizable.

X is **locally metrizable** if each point has a neighbourhood that is metrizable in the subspace topology.

Theorem 6.14 (42.1, Smirnov metrization theorem)

X is metrizable iff it is a paracompact Hausdorff space that is locally metrizable.

Proof. ($\Rightarrow$) Theorem 6.9.

($\Leftarrow$) X is regular (Theorem 6.6); we find a countably locally finite basis and apply Theorem 6.5. Cover X by open metrizable sets and take a locally finite open refinement $\mathcal{C}$ covering X; each $C \in \mathcal{C}$ has a metric d_C, and the balls $B_C(x,\epsilon) = \{y \in C : d_C(x,y) < \epsilon\}$ are open in X. For each m, let $\mathcal{A}_m$ be the covering by all balls $B_C(x, 1/m)$ ($C \in \mathcal{C}$, $x \in C$), and $\mathcal{D}_m$ a locally finite open refinement covering X. Then $\mathcal{D} = \bigcup \mathcal{D}_m$ is countably locally finite. Basis: let $x \in U$ open. Only finitely many $C_1, \dots, C_k \in \mathcal{C}$ contain x; choose ϵ_i with $B_{C_i}(x,\epsilon_i) \subset U \cap C_i$ and m with $2/m < \min \epsilon_i$. Choose $D \in \mathcal{D}_m$ with $x \in D$; then $D \subset B_C(y, 1/m)$ for some C and y, and $x \in C$, so $C = C_i$ for some i. By the triangle inequality $D \subset B_{C_i}(x, 2/m) \subset U$. ∎